Yonghao Shi

Ph.D. Student in Computer Science, Simon Fraser University

Human-Computer Interaction | Smart Environments | Computational Materials | Applied AI

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PROFILE

My research lies at the intersection of Human-Computer Interaction, smart environments, computational materials, sensing systems, and applied AI. I design hardware-enabled interactive systems that turn everyday materials, especially wood panels, furniture, and room-scale structures, into sensing and actuation interfaces. My current work focuses on sustainable computational wood, battery-free wireless sensing, RFID-augmented environments, low-barrier fabrication, and user-centered evaluation. Before joining SFU, I worked as a software engineer and project manager, which shaped my ability to build reliable systems and manage research prototypes end to end.

SCHOLARSHIP

Award	SFU PhD Research Scholarship, Simon Fraser University.
Award	Graduate Fellowship, Simon Fraser University. Multiple awards during Ph.D. study.

PUBLICATION

Published papers

[C4] **3Duino: A Low-Barrier Platform for Prototyping Interactive 3D-Printed Devices.**
Yonghao Shi, Zejia Cai, Liuyi Chen, Yuning Su, Liang He, Xing-Dong Yang, and Te-Yen Wu.
The 10th Annual ACM Symposium on Computational Fabrication (SCF 2025). DOI: 10.1145/3745778.3766649

[C3] **Woodowel: Electrode Isolation for Electromagnetic Shielding in Triboelectric Plywood Sensors.**
Yonghao Shi, Chenzheng Li, Yuning Su, Xing-Dong Yang, and Te-Yen Wu.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2024). DOI: 10.1145/3613904.3642304

[C2] **Tagnoo: Enabling Smart Room-Scale Environments with RFID-Augmented Plywood.**
Yuning Su, Tingyu Zhang, Jiuen Feng, **Yonghao Shi**, Xing-Dong Yang, and Te-Yen Wu.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2024). DOI: 10.1145/3613904.3642356

[C1] **Laser-Powered Vibrotactile Rendering.**
Yuning Su, **Yonghao Shi**, Da-Yuan Huang, and Xing-Dong Yang.
Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct 2023). DOI: 10.1145/3586182.3615795

Manuscripts under review

- Lumberina: A Battery-Free, Wireless Smart Wood Panel with Concealed Dot-Matrix Display.
- EcoVibWood: A Sustainable Vibration-Sensing Plywood Panel with Eco-Friendly Sensing Layers and a Self-Isolating Electrode Switching Mechanism.
- Exploring Tangible Devices for Low-Effort, Participant-Controlled Data Input.

EDUCATION

2022 - Present	**Ph.D. in Computer Science**, Simon Fraser University, Burnaby, Canada. Research: Human-Computer Interaction, smart environments, and computational materials.
2017 - 2018	**M.Sc. in Computer Science**, University of Plymouth, United Kingdom. Thesis: Data Science and Business Analysis.
2016 - 2017	**B.Sc. in Electrical and Electronic Engineering**, University of Plymouth, United Kingdom.
2013 - 2016	**B.Sc. in Electrical and Electronic Engineering**, Beihang University, China.

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